Aviation Weather Terminal Aerodrome Forecast (TAF) Specification & Verifica

Authors / Researchers:
Research Team R26-IT-132 | Department of Meteorology Sri Lanka & SLIIT CoEAI

EXECUTIVE ABSTRACT & SYSTEM SCOPE

Standard operational guideline and technical specification for Terminal Aerodrome Forecast (TAF) decoding, verification criteria, and ICAO Annex 3 compliance. Provides the meteorological benchmark and threshold values utilized by the automated forecast verification engine.

1. Purpose and Scope of TAF Verification

Terminal Aerodrome Forecasts (TAF) represent concise statements of expected meteorological conditions at an airport for a specified period (typically 24 to 30 hours).

Automated verification compares issued TAF reports against valid METAR ground observations at 30-minute intervals to evaluate operational forecast fidelity and identify systematic model biases.

2. ICAO Annex 3 Operational Thresholds for VCBI

" Wind Direction & Speed: Significant change defined as direction variation >= 30 deg when mean speed before or after change is >= 20 kt, or speed change >= 10 kt.

" Visibility Thresholds: Verified against category milestones: 800m, 1500m, 3000m, and 5000m for precision instrument approaches.

" Cloud Base Height: Significant changes monitored at 100ft, 200ft, 500ft, 1000ft, and 1500ft ceiling minima.

" Significant Weather: Onset and cessation of convective storms, heavy tropical rain (RA), and mist/fog (FG/BR).

3. Verification Metrics & Scoring Algorithms

The verification pipeline employs categorical scoring (Hit Rate, False Alarm Ratio, Critical Success Index) combined with continuous error metrics (MAE, RMSE) across each forecast validity window.